
The Gaussian Toolkit

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Notation: ■ x = the unobserved (latent) sub-vector ■ y = the observed sub-vector

The object of study: a partitioned Gaussian

Jointly Gaussian pair (x, y)

Let $z = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^{d_x+d_y}$ be Gaussian,

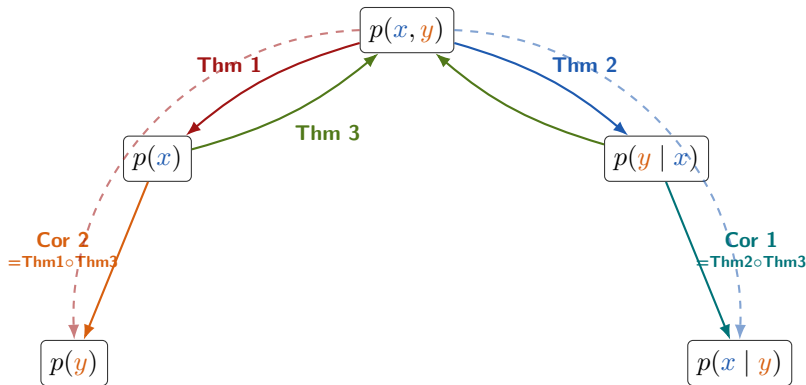
$$z \sim \mathcal{N}(\mu, \Sigma), \quad \mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}, \quad \Sigma_{yx} = \Sigma_{xy}^\top.$$

$x \in \mathbb{R}^{d_x}$ and $y \in \mathbb{R}^{d_y}$ may have any dimensions.

- **Roles.** x is the quantity we wish to infer (latent state, parameters); y is the quantity we measure (data). The partition is a modelling choice — the joint law is symmetric in x and y .
- From this single object we derive **five** standard distributions: the joint, the two *marginals*, and the two *conditionals*.

Examples: $(x, y) =$ (weights, outputs) in regression; (true state, sensor reading) in tracking; (clean signal, noisy signal) in denoising.

One joint distribution, five representations (Figure 3)



The nodes are the **joint** $p(x, y)$, the two **marginals** $p(x), p(y)$, and the two **conditionals** $p(y | x), p(x | y)$ of the *same* Gaussian. Each edge is an exact operation: marginalization (Thm 1), conditioning (Thm 2), assembly of the joint from a prior and a forward model (Thm 3); the corollaries are compositions.

Five representations, five inference queries

Representation	Query it answers	Required inputs	Typical use
joint $p(\mathbf{x}, \mathbf{y})$	How do $\mathbf{x}$, $\mathbf{y}$ covary?	μ, Σ (all blocks)	complete description
marginal $p(\mathbf{x})$	Law of $\mathbf{x}$ alone?	the joint (or a prior)	prior / baseline belief
conditional $p(\mathbf{y} \mathbf{x})$	Law of $\mathbf{y}$ for fixed $\mathbf{x}$?	the measurement model	likelihood; simulation
marginal $p(\mathbf{y})$	Law of $\mathbf{y}$ alone?	prior $p(\mathbf{x})$ + model	predictive; evidence
conditional $p(\mathbf{x} \mathbf{y})$	Law of $\mathbf{x}$ given observed $\mathbf{y}$?	prior + model + $\mathbf{y}$	posterior; inference

One typically *possesses* some representations (a prior and a forward model) and *requires* another (the posterior or the predictive). Figure 3 specifies the operation that connects them.

Theorems 1–2, and the Schur complement

Theorem 1 (marginalization)

$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{y}) d\mathbf{y} = \mathcal{N}(\mathbf{x}; \mu_x, \Sigma_{xx})$ — read the mean/covariance block directly.

Theorem 2 (conditioning)

$p(\mathbf{x} \mid \mathbf{y}) = \mathcal{N}(\mathbf{x}; \mu_{x|y}, \Sigma_{x|y})$ with

$$\mu_{x|y} = \mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (\mathbf{y} - \mu_y), \quad \Sigma_{x|y} = \Sigma_{xx} - \Sigma_{xy} \Sigma_{yy}^{-1} \Sigma_{yx}.$$

Schur complement

For a block matrix $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ with D invertible, the *Schur complement* of D in M is $M/D := A - BD^{-1}C$. Thus the conditional covariance is $\Sigma_{x|y} = \Sigma / \Sigma_{yy}$.

Note $\Sigma_{x|y}$ does *not* depend on the observed value $\mathbf{y}$ — only the conditional *mean* does. Conditioning reduces covariance.

The two conditionals: forward and inverse directions

The same relationship, factored two ways.

$p(y | x)$ — forward model / likelihood

The law of the observation given the latent value. It *specifies the model* (the measurement / data-generating process). Viewed as a function of x for observed y , it is the **likelihood** $\ell(x) = p(y | x)$.

$p(x | y)$ — inverse / posterior

The law of the latent quantity given the observation. It is the **inference target**: parameter or state estimation, denoising, or solving an inverse problem.

The forward direction is assumed (it is the model); the inverse direction is computed from it by Theorem 2 (equivalently Corollary 1). In calibration: forward = instrument specification; inverse = recovery of the true quantity from a reading.

The marginal $p(\mathbf{y})$: the predictive distribution

$$p(\mathbf{y}) = \int p(\mathbf{y} | \mathbf{x}) p(\mathbf{x}) d\mathbf{x} = \mathcal{N}(\mathbf{y}; \mu_{\mathbf{y}}, \Sigma_{\mathbf{y}}) \quad (\text{Corollary 2}).$$

It gives the law of the observation with the latent $\mathbf{x}$ integrated out.

- **Forecasting**: the distribution of observations the model predicts.
- **Model evidence**: $p(\mathbf{y})$ is the normalizing constant in Bayes' rule, used to compare or select models.
- **Anomaly detection**: an observed $\mathbf{y}$ with very small $p(\mathbf{y})$ is inconsistent with the model.

In Bayesian linear regression, using the prior $p(\mathbf{x})$ gives the *prior predictive*; using the posterior $p(\mathbf{x} | \text{data})$ gives the *posterior predictive*.

Bayes' rule: the origin of the names

The joint factors two ways, $p(\textcolor{blue}{x}, \textcolor{brown}{y}) = p(\textcolor{brown}{y} \mid \textcolor{blue}{x})p(\textcolor{blue}{x}) = p(\textcolor{blue}{x} \mid \textcolor{brown}{y})p(\textcolor{brown}{y})$, which rearranges to

$$\underbrace{p(\textcolor{blue}{x} \mid \textcolor{brown}{y})}_{\text{posterior}} = \frac{\overbrace{p(\textcolor{brown}{y} \mid \textcolor{blue}{x})}^{\text{likelihood}} \overbrace{p(\textcolor{blue}{x})}^{\text{prior}}}{\underbrace{p(\textcolor{brown}{y})}_{\text{evidence}}}.$$

- **Posterior** $p(\textcolor{blue}{x} \mid \textcolor{brown}{y})$: the law of $\textcolor{blue}{x}$ *after* conditioning on the observation — the prior updated by the data.
- **Predictive** $p(\textcolor{brown}{y})$: a distribution over the *observable*, obtained by marginalizing $\textcolor{blue}{x}$; it also serves as the evidence (denominator).

Dimensions: the toolkit is not restricted to $d = 2$

The bivariate contour plots are two-dimensional only for visualization. All results hold for arbitrary block dimensions d_x, d_y , using the same block mean and covariance and the Schur complement of Theorem 2.

Bayesian linear regression as a special case

The weights are the latent vector, $\mathbf{x} = \mathbf{w} \in \mathbb{R}^D$. The targets form an N -dimensional vector $\mathbf{y} \in \mathbb{R}^N$ that *stacks the N scalar outputs $y_1, \dots, y_N$* (one per training example). The joint $(\mathbf{w}, \mathbf{y})$ lives in $\mathbb{R}^{D+N}$.

Terminology check: $\mathbf{y} \in \mathbb{R}^N$ is a vector of N scalar outputs, not a single N -dimensional (multi-output) response. Multi-output regression would instead give each example a vector target $\mathbf{y}_n \in \mathbb{R}^k$.

Affine–Gaussian forward model

$$\mathbf{y} = A\mathbf{x} + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Sigma_{y|x}), \quad \varepsilon \perp \mathbf{x}, \quad \mathbf{x} \sim \mathcal{N}(\mu_x, \Sigma_x).$$

$A \in \mathbb{R}^{d_y \times d_x}$ is a linear map (gain / design matrix), $b \in \mathbb{R}^{d_y}$ an offset, and ε additive Gaussian measurement noise.

- **When it applies:** the observation is (approximately) an affine function of the latent quantity with additive Gaussian noise. Then all five representations are Gaussian and available in closed form.
- **Instances:** a sensor with gain A and offset b ; linear regression $\mathbf{y} = \mathbf{X}\mathbf{w} + \varepsilon$ ($A = \mathbf{X}$, $\mathbf{x} = \mathbf{w}$); the Kalman measurement equation; factor analysis.

With a Gaussian prior $p(\mathbf{x})$, this model **assembles the joint** $p(\mathbf{x}, \mathbf{y})$ (Theorem 3, next). “Assemble” is meant precisely: Theorem 3 is the unique operation in Figure 3 whose output is the joint node; every other representation is then obtained from the joint.

From a regression equation to a probability model

Consider the observation model

$$\mathbf{y} = A\mathbf{x} + \mathbf{b} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \boldsymbol{\varepsilon} \perp \mathbf{x}.$$

Here, A and $\mathbf{b}$ are treated as fixed parameters.

Fix the input $\mathbf{x}$. Then $A\mathbf{x} + \mathbf{b}$ is a constant, but $\mathbf{y}$ remains random because of the noise.

- Adding a constant to Gaussian noise preserves Gaussianity.
- The constant shifts the mean:

$$\mathbb{E}[\mathbf{y} \mid \mathbf{x}] = A\mathbf{x} + \mathbf{b}.$$

- The constant does not change the covariance:

$$\text{Cov}(\mathbf{y} \mid \mathbf{x}) = \Sigma.$$

Thus, the same model can be written as

$$\boxed{\mathbf{y} \mid \mathbf{x} \sim \mathcal{N}(A\mathbf{x} + \mathbf{b}, \Sigma).}$$

Interpretation: the regression function gives the centre of the output distribution; the noise covariance describes the uncertainty around it.

Theorem 3: assembling the joint

Theorem 3 (affine–Gaussian $\Rightarrow$ jointly Gaussian)

Given $p(\mathbf{x}) = \mathcal{N}(\mu_x, \Sigma_x)$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(A\mathbf{x} + b, \Sigma_{y|x})$, the pair is jointly Gaussian:

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_x \\ A\mu_x + b \end{pmatrix}, R\right), \quad R = \begin{pmatrix} \Sigma_x & \Sigma_x A^\top \\ A\Sigma_x & \Sigma_{y|x} + A\Sigma_x A^\top \end{pmatrix}.$$

- Theorem 3 is the **assembly step**: it maps the prior and the forward model to the joint $p(\mathbf{x}, \mathbf{y})$. This is the precise meaning of “constructing the joint” in Figure 3.
- The measurement noise enters only the **(y, y) block**, as $\Sigma_{y|x} + A\Sigma_x A^\top$: sensor noise *plus* prior uncertainty propagated through A .
- Applying Theorem 1 or Theorem 2 to R now yields the predictive and the posterior — the two corollaries.

Corollary 2: the predictive distribution (Theorem 3 then Theorem 1)

Corollary 2 (marginal of y)

Marginalizing x out of the joint gives

$$p(y) = \mathcal{N}(y; \underbrace{A\mu_x + b}_{\mu_y}, \underbrace{\Sigma_{y|x} + A\Sigma_x A^\top}_{\Sigma_y}).$$

Derivation: read the (y) -blocks of R from Theorem 3, i.e. apply Theorem 1. The variance has two sources: measurement noise $\Sigma_{y|x}$ and propagated prior uncertainty $A\Sigma_x A^\top$.

Applications of Corollary 2

- Prior-predictive **forecasting** of observations before any data are seen.
- **Model evidence** $p(y)$ for Bayesian model comparison and hyperparameter selection.
- **Anomaly / novelty detection** and **optimal experimental design**.

Corollary 1: the posterior (Theorem 3 then Theorem 2)

Corollary 1 (posterior of x given y)

Conditioning the joint on y gives $p(x | y) = \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$,

$$\Sigma_{x|y} = (\Sigma_x^{-1} + A^\top \Sigma_{y|x}^{-1} A)^{-1}, \quad \mu_{x|y} = \Sigma_{x|y} (\Sigma_x^{-1} \mu_x + A^\top \Sigma_{y|x}^{-1} (y - b)).$$

Derivation: apply Theorem 2 (the Schur complement) to R ; the information form above follows by the matrix-inversion lemma. Prior precision Σ_x^{-1} and data precision $A^\top \Sigma_{y|x}^{-1} A$ add.

Applications of Corollary 1

- **Bayesian parameter estimation:** with $x = w$, $A = X$, this is the BLR weight posterior $\mathcal{N}(\mu_N, \Sigma_N)$.
- **State estimation / sensor fusion:** the Kalman-filter measurement update; combining GPS and inertial readings.
- **Denoising and inverse problems:** recovering a latent signal from noisy observations.

Worked example: a scalar sensor

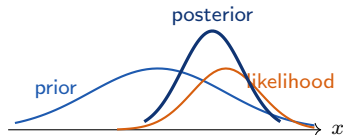
Unit-gain sensor: latent $x \sim \mathcal{N}(0, 1)$, reading $y = x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.5^2)$
($A = 1$, $b = 0$, $\Sigma_{y|x} = 0.25$).

Applying the corollaries

Likelihood (given): $p(y | x) = \mathcal{N}(x, 0.25)$.

Predictive (Corollary 2): $p(y) = \mathcal{N}(0, 1 + 0.25) = \mathcal{N}(0, 1.25)$.

Posterior (Corollary 1), on observing $y = 1$: $\Sigma_{x|y} = (1 + 1/0.25)^{-1} = 0.2$, $\mu_{x|y} = 0.2 \cdot (0 + 4 \cdot 1) = 0.8$,
so $p(x | y=1) = \mathcal{N}(0.8, 0.2)$.



With gain $A = 2$ and offset $b = 1$, Corollary 2 gives $p(y) = \mathcal{N}(1, 4.25)$. Replacing $x \rightarrow w$ and $A \rightarrow X$ yields the BLR predictive and weight posterior.

- A partitioned Gaussian has five standard representations: the **joint**, two **marginals**, and two **conditionals**. Figure 3 lists the exact operations connecting them.
- **Theorem 1** marginalizes; **Theorem 2** conditions (its covariance is the **Schur complement** Σ/Σ_{yy}); **Theorem 3** assembles the joint from a prior and a forward model.
- $p(\mathbf{y} \mid \mathbf{x})$ is the forward model / likelihood; $p(\mathbf{x} \mid \mathbf{y})$ is the posterior; $p(\mathbf{y})$ is the predictive (and the evidence).
- For $\mathbf{y} = A\mathbf{x} + b + \varepsilon$: the affine-transformation law gives $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(A\mathbf{x} + b, \Sigma_{y|x})$; **Corollary 2** then yields the predictive and **Corollary 1** the posterior.
- All results hold for arbitrary dimensions; Bayesian linear regression is the case $\mathbf{x} = \mathbf{w} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^N$.